

Yunus Serhat Bıçakçı, PhD

Geospatial AI | Vision-Language Models | Remote Sensing | Spatial Data Science

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Research Profile

Assistant Professor and researcher working in geospatial AI, vision-language models, remote sensing and spatial data science. My research explores how artificial intelligence and geographic information can help us understand environmental and urban questions. I work with colleagues in Türkiye and the UK on research projects, publications and open research software.

Research Highlights

Research leadership and funding. Principal Investigator on an Istanbul street imagery project awarded TRY 546,500 and on SokakPuan, awarded TRY 100,000. Recipient of a GBP 49,200 TÜBİTAK international postdoctoral fellowship.

Research recognition. First place in both phases of the WHISPERS 2025 ISASeg Challenge. Coauthor of the paper that received the AGILE 2026 Best Short Paper Award.

Open research and international service. Author of GeoAI-VLM and sppt, two open source research tools archived with DOIs. International Committee Chair for GIScience 2027.

Academic and Research Appointments

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| 2026–present | Assistant Professor
Department of Artificial Intelligence and Machine Learning, Faculty of Applied Sciences, Marmara University , Istanbul, Türkiye.
Conduct research and teach in artificial intelligence and data science. Contribute to curriculum development, research projects and academic collaboration. |
| 2024–present | Research Software Engineering Scientist / Affiliate
Geospatial Data Science Group, School of Geographical and Earth Sciences, University of Glasgow , UK.
Work as part of the Geospatial Data Science Group on collaborative research, manuscript preparation, conference presentations and joint project development. |
| 2023–2024 | Postdoctoral Researcher
Jill Dando Institute of Security and Crime Science, University College London , UK.
Conducted postdoctoral research in geospatial analysis and data science. Collaborated on publications and presented research at academic events. |
| 2022–2023 | Assistant Professor
Vocational School of Social Sciences, Marmara University , Istanbul, Türkiye.
Taught undergraduate courses and contributed to research, curriculum development and departmental activities. |
| 2021–2022 | Research Assistant, PhD
Vocational School of Social Sciences, Marmara University , Istanbul, Türkiye.
Supported teaching, research and departmental activities. |

Research Funding and Fellowships

- 2026–2027 **AI-Based Analysis of Istanbul Street-Level Imagery and Creation of an Open Dataset**
Principal Investigator, TRY 546,500 awarded
Research Universities Support Program, project 12127, 12 months.
Lead a GeoAI project to analyse approximately three million street images of Istanbul and develop an open dataset for urban and environmental research.
- 2026 **SokakPuan**
Principal Investigator, TRY 100,000 awarded
STEPS Open Innovation Call (Sokaklar Dönüşüyor), supported by EIT Urban Mobility.
Lead the development and municipal piloting of a platform that uses imagery, GIS, artificial intelligence and citizen feedback to evaluate streets and public spaces.
- 2023–2025 **TÜBİTAK 2219 International Postdoctoral Research Fellowship**
Fellowship recipient, GBP 49,200 awarded over two years
Competitive TÜBİTAK fellowship supporting postdoctoral research at University College London on artificial intelligence, crime mapping and urban data science.
- 2024 **Multi-Lingual and Multi-Modal Location Information Extraction (Multi-LM)**
Research Associate / Researcher, total project funding GBP 394,911
The Alan Turing Institute, grant 317199-2.
Joined the project in 2024. Contributed to application development and geolocation research at Glasgow and coauthored a paper from the project.

Awards and Recognition

- 2026 **Best Short Paper Award, AGILE 2026**
Coauthor of the short paper on semantic neighbourhoods in London's Airbnb properties that received the award at the 29th AGILE Conference, Tartu, Estonia.
- 2025 **1st Place, WHISPERS ISASeg Challenge, Phase 1 and Phase 2**
First in both phases using multimodal ensembles to map impervious surfaces from Sentinel-1 and Sentinel-2 imagery.
- 2024 **Global Talent Endorsement, The British Academy**
Endorsed under the UK's Global Talent visa scheme for research promise and academic contribution.

Academic Leadership and Professional Service

- 2026–present **International Committee Chair, GIScience 2027**
Fourteenth International Conference on Geographic Information Science.
- 2023–2025 **Vice Head of Department, Marmara University**
Department of Office Technologies and Data Management, Vocational School of Social Sciences.
- 2023–present **Delegate, Environmental Criminology and Crime Analysis (ECCA)**
- 2022 **Editorial Board / Co-Editor**
The Social Research Journal of Marmara University.
- 2026 **Journal Refereeing**
- 2021–2022 *International Journal of Environment and Geoinformatics* (2026) and *International Humanities and Social Science Review* (2021–2022).

Education

- 2014–2021 **PhD in Applied Informatics, Istanbul Technical University**, Türkiye
Geographical Information Technologies Program.
Dissertation. *Spatio-Temporal Analysis of Applied Electronic Monitoring on Parolees of Mobility: A Case Study in Istanbul*. Thesis
Supervised by Prof. Dursun Zafer Şeker.
- 2011–2013 **MSc in Information Systems, Sakarya University**, Türkiye
Management Information Systems Program. Supervised by Prof. Orhan Torkul.
- 2007–2011 **BEng, Ankara University**, Türkiye
- 2007–2011 **BA, Anadolu University**, Türkiye

Peer-Reviewed Journal Articles

- 2026 **Performance Comparison of Multimodal Vision-Language Models in Classifying Turkish Dishes**
Yunus Serhat Bıçakçı.
Sakarya University Journal of Computer and Information Sciences, 9(1), 119–133. DOI
- 2025 **Comparative Assessment of CNN and Transformer U-Nets in Multiple Sclerosis Lesion Segmentation**
Beytullah Sarıca, **Yunus Serhat Bıçakçı**, Dursun Zafer Şeker.
International Journal of Imaging Systems and Technology, 35(4), e70146. DOI
- 2023 **ATTransUNet: Semantic Segmentation Model for Building Segmentation from Aerial Image and Laser Data**
Yunus Serhat Bıçakçı, Beytullah Sarıca.
Nordic Machine Intelligence, 2, 7–9. DOI
- 2020 **Location-Based Analyses for Electronic Monitoring of Parolees**
Yunus Serhat Bıçakçı, Dursun Zafer Şeker, Hande Demirel.
ISPRS International Journal of Geo-Information, 9(5), 296. DOI
- 2019 **Spora Yönelik Tutum İle Yaşam Doyumu Arasındaki İlişki: Marmara Üniversitesi Sosyal Bilimler Meslek Yüksekokulu Öğrencileri Örneği**
Çağdaş Caz, **Yunus Serhat Bıçakçı**, Fatma Firuzan Nakipoğlu.
Gümüşhane Üniversitesi Sosyal Bilimler Enstitüsü Elektronik Dergisi, 10(3), 484–491. Full text
- 2017 **Spatio-Temporal Analyses to Estimate Speed Information for Transport Forecasting and Scenario Testing**
Yunus Serhat Bıçakçı, Beytullah Sarıca, Mete Ercan Pakdil, Batuhan Yazırlı, Hande Demirel.
Fresenius Environmental Bulletin, 26(1), 100–106. Publication link

Peer-Reviewed Conference Papers and Proceedings

- 2026 **A Graph-Based Community Detection Approach for Identifying the Semantic Neighbourhoods within London's Airbnb Properties**
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Yu Wang, Ana Basiri, Meiliu Wu.
AGILE: GIScience Series, 7, 41. Peer-reviewed short paper, 29th AGILE Conference, 16–19 June, Tartu, Estonia. DOI
Best Short Paper Award.
- 2026 **Mapping the Semantics of the Street: A VLM-Driven Geospatial Analysis in Fatih, Istanbul**
Yunus Serhat Bıçakçı, Joseph Shingleton, Yu Wang, Anahid Basiri.
Peer-reviewed full paper, 1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026), 3–6 June, Ghent, Belgium. DOI
- 2026 **The Invisible City: Not All Streets Are Seen Equally**
Yu Wang, Joseph Shingleton, **Yunus Serhat Bıçakçı**, Anahid Basiri, James Ackland.
Peer-reviewed full paper, GeoAI 2026, 3–6 June, Ghent, Belgium. DOI
- 2026 **Emergent Spatio-Semantic Structure in Large Language Model Embedding Spaces**
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Yu Wang, Anahid Basiri.
Peer-reviewed paper presented orally at GeoAI 2026, 3–6 June, Ghent, Belgium. Associated [Earth-ArXiv preprint](#).

Book Chapters

- 2026 **Towards GeoAI Foundation Models: A Multimodal Learning Framework with Geospatial Intelligence**
Meiliu Wu, Qunying Huang, Anahid Basiri, **Yunus Serhat Bıçakçı**, Joseph Shingleton.
In *Geography According to Foundation Models*, Frontiers in Artificial Intelligence and Applications, vol. 422, IOS Press, pp. 28–50. DOI
- 2025 **Big Data and Social Media Analytics: Opportunities for Interactive Crime Mapping**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
In M. Leitner (ed.), *New Research in Crime Modeling and Mapping Using Geospatial Technologies*, Geotechnologies and the Environment, Springer, pp. 3–25. DOI
- 2025 **Yapay Zekanın Sokak Seviyesi Görüntü Analizinde Kullanımı**
Yunus Serhat Bıçakçı.
In M. Tokmak (ed.), *Türkiye ve Dünyada Yönetim Bilişim Sistemleri*, YAZ Yayınları, Afyonkarahisar, pp. 77–93.
- 2022 **Turizm ve Seyahat Bürolarında Yapay Zeka Uygulamaları**
Yunus Serhat Bıçakçı, Bora Aksu.
In D. Semiz Çelik (ed.), *Dijital Dönüşümün Turizme Yansımaları*, Beta Yayınları, İstanbul.

Research Software and Interactive Applications

Interactive AI and Geospatial Applications

Developed Gradio and Streamlit applications on Hugging Face Spaces, including [Location Predictor](#) and [Crime-Map](#). Practical front-end and back-end development combining user-facing interfaces, Python application logic, model inference pipelines, APIs and geospatial data processing in end-to-end research prototypes.

2026 **GeoAI-VLM: Geospatial Vision-Language Model Analysis**

Yunus Serhat Bıçakçı. DOI | [GitHub](#)

Open source software for downloading Mapillary imagery by location and generating descriptions with vision-language models. Release v0.3 archived on Zenodo.

2026 **sppt: Spatial Point Pattern Test for Aggregated Data (Python)**

Yunus Serhat Bıçakçı. DOI | [GitHub](#)

Open source Python implementation using bootstrap resampling and S-Index metrics to compare spatial distributions of aggregated count data. Based on Martin A. Andresen's original R implementation.

Research Methods and Technical Expertise

Geospatial and environmental analysis. GIS, spatial analysis, Sentinel-1 and Sentinel-2, integration of radar and optical imagery, street imagery, field imagery, environmental mapping, QGIS, ArcGIS, GeoPandas and PostGIS.

Computer vision and multimodal AI. Semantic segmentation, object detection, vision transformers, vision-language models, geospatial foundation models, dataset curation and multimodal evaluation.

Foundation-model fine-tuning and application. Hands-on adaptation and fine-tuning of pretrained models, including CLIP, SigLIP and vision-language models, for image classification, geospatial analysis and street-level imagery. Embedding-based workflows, multimodal evaluation, RAG-based geolocation and integration into research pipelines.

Programming and data. Python, R, SQL, Bash, NumPy, Pandas, OpenCV, DuckDB and data visualisation.

Front-end and back-end development. Gradio and Streamlit interfaces, Hugging Face Spaces, Python application development, APIs, model inference pipelines and geospatial data processing for interactive AI research prototypes.

AI systems and reproducible software. Retrieval augmented generation, LlamaIndex, LangGraph, FAISS, Transformers, vLLM, model quantisation and serving, Docker, Git, CI/CD, experiment tracking, Google Cloud and Microsoft Azure.

Public Talks and Outreach

2022 **TensorFlow Basics**, Google DSC Bandırma University, December. [Video](#)

Intro to Machine Learning, Google DSC Işık University, November. [Video](#)

Manuscripts Under Review and Revision

- 2026 **AlphaEarth embeddings match conventional Earth observation for street-level greenery**
Yunus Serhat Bıçakçı.
Photogrammetric Engineering & Remote Sensing. Major revision requested. Revision in preparation.
- 2026 **Can generative vision models measure micro-place environments? An econometric assessment...**
Journal of Quantitative Criminology. Under peer review.
- 2026 **When Does Detector Fusion Fail? A Systematic Benchmark of YOLO and DETR Ensembles for Military Vehicle Detection in UAV Imagery**
Yunus Serhat Bıçakçı.
Signal, Image and Video Processing. Under editorial assessment (with editor).
- 2026 **Impervious Surface Segmentation from Multisource Satellite Imagery Using Early-Fusion Deep Ensembles**
Yunus Serhat Bıçakçı.
International Journal of Environment and Geoinformatics. Under peer review.
- 2026 **Evaluating Explanation Integrity in a Vision Language Model for Urban Perception**
Joseph Shingleton, **Yunus Serhat Bıçakçı**, Meiliu Wu, Yu Wang, Ana Basiri.
Transactions in GIS. Revised manuscript resubmitted following major revision and returned to peer review.
- 2026 **A use-conditioned due-diligence protocol for open AI-derived geospatial products: Application to Fields of The World in Türkiye**
Yunus Serhat Bıçakçı.
Discover Geoscience. Under editorial assessment (with editor).

Preprints

- 2026 **Hybrid CNN-Transformer Ensemble for Enhanced Tank Detection in Aerial Imagery**
Yunus Serhat Bıçakçı.
Research Square preprint. DOI
- 2025 **Street-Level Geolocalization Using Multimodal Large Language Models and Retrieval-Augmented Generation**
Yunus Serhat Bıçakçı, Joseph Shingleton, Anahid Basiri.
arXiv:2509.01341 [cs.CV]. DOI

Selected Conference Presentations and Abstracts

- 2025 **Digitalising Probation with Geospatial AI: Street-Level Human Perception in Istanbul's Fatih District**
Yunus Serhat Bıçakçı, Alina Ristea.
Turkish Probation System 20th Year International Congress, 1–3 December, Antalya, Türkiye. Conference summary.
- 2025 **Multimodal Semantic Segmentation for Impervious Surface Mapping**
Yunus Serhat Bıçakçı.
15th Workshop on Hyperspectral Image and Signal Processing: Evolution of Remote Sensing (WHISPERS), 12–14 November, Bellaterra, Barcelona, Spain.
- 2023 **Using Big Data, Artificial Intelligence and Cloud Technologies for Interactive Crime Mapping**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Stockholm Criminology Symposium, 12–14 June, Stockholm, Sweden. [Programme](#)
- 2023 **Application of Natural Language Processing Techniques for Crime Detection**
Yunus Serhat Bıçakçı, Alina Ristea, Kate Bowers.
Environmental Criminology and Crime Analysis (ECCA), 14–16 June, Stockholm, Sweden. [Programme](#)
- 2022 **Artificial Intelligence Applications for Social Municipality: The Case of Chatbot**
Yunus Serhat Bıçakçı.
International VI. Civil Strategy Symposium, 8–9 December, Istanbul, Türkiye. [Proceedings](#)

Teaching and Mentorship

- 2026–2027 **Marmara University, Department of Artificial Intelligence and Machine Learning**
Scheduled for autumn 2026. Data Structures and Career Planning.
- 2025–2026 **Marmara University, Vocational School of Social Sciences**
Information and Communication Technologies (autumn and spring), Office Applications, Media Literacy and Management Information Systems.
- 2024–2025 **Marmara University, Vocational School of Social Sciences**
Information and Communication Technologies, Management Information Systems, Telephone Communication and Time Management.
- 2022–2023 **Marmara University, Vocational School of Social Sciences**
Office Applications, Computer I and Media Literacy.
- 2023–2026 **Mentor, inzva Mentorship Program**
Mentored six undergraduate students in their third or fourth year across three annual cohorts. Provided guidance on research planning, technical skills, academic development and careers in computer science and AI.